1) Programare imperativă:

1.1) SEMANTICA OPERAȚIONALĂ:

1) p = if $x \leq 7$ then $z := x$ else $z := y$

$\sigma(x) = 0$; $\sigma(y) = 1$; $\sigma(z) = 7$

Sol.: (B.S.)

$(p, \sigma) \Downarrow \sigma_{z \mapsto 0}$

$e_\sigma(x \leq 7) = 1$ ($0 \leq 7$ - adevărat) $(z := x, \sigma) \Downarrow \sigma_{z \mapsto 0}$

$(p, \sigma) \Downarrow \sigma_{z \mapsto 0}$ (regula 1 de la if)

(S.S)

$(p, \sigma) \rightarrow (z := x, \sigma) \rightarrow (skip, \sigma_{z \mapsto e_\sigma(x)}) \rightarrow (skip, \sigma_{z \mapsto 0})$

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2) p = while $4 \leq x$ do $x : x - y$

$\sigma(x) = 6$; $\sigma(y) = 3$

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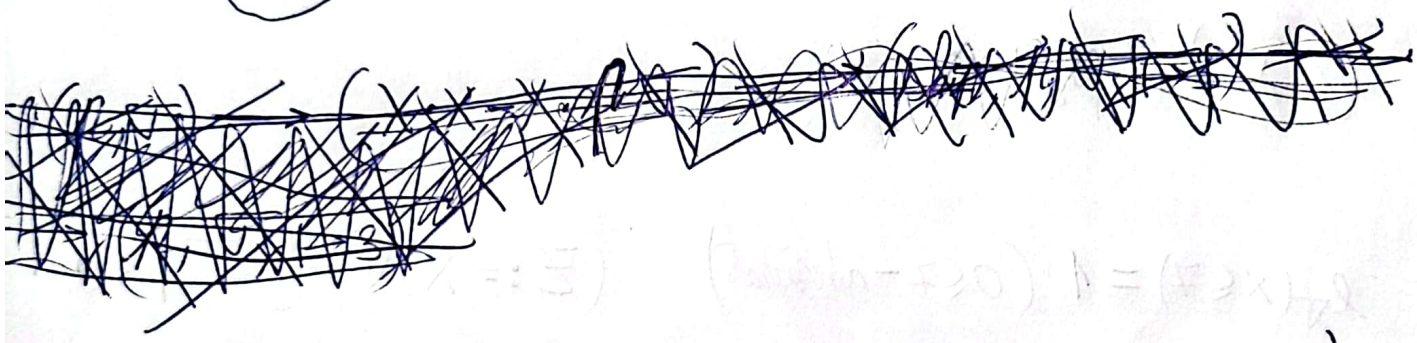
Sol.: (B.S.)

$\mathcal{E}_{\sigma}(4 \leq X) = 1$ ($4 \leq 6$ – adevărat)

$(X : X - Y, \sigma) \Downarrow \sigma_{X \mapsto 3}$

$(p, \sigma) \Downarrow \sigma_{X \mapsto 3}$ (regula 2 while)

(S.S)



$(p, \sigma) \rightarrow (\text{if } 4 \leq X \text{ then } X := X - Y; p \text{ else skip}, \sigma) \rightarrow$

$\rightarrow (X : X - Y, p, \sigma) \rightarrow (\text{skip}, p, \sigma_{X \mapsto 3}) \rightarrow (p, \sigma_{X \mapsto 3}) \rightarrow$

$\rightarrow (\text{if } 4 \leq X \text{ then } X : X - Y; p \text{ else skip}, \sigma_{X \mapsto 3}) \rightarrow$

$\rightarrow (\text{skip}, \sigma_{X \mapsto 3})$

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3) $p = \text{while } \neg(Y = 0) \text{ do } (Y : Y - 1; X : 2 * X)$

$\sigma(X) = 1 ; \sigma(Y) = 3$

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B.S.

$e_\sigma(\neg(Y=0)) = 1$

$((Y:=Y-1, X:=X*2), \sigma_{X\mapsto 1, Y\mapsto 3}) \Downarrow_{X\mapsto 2, Y\mapsto 2}$

$(n, \sigma_{X\mapsto 1, Y\mapsto 3}) \Downarrow \sigma_{X\mapsto 2, Y\mapsto 2}$

$e_\sigma(\neg(Y=0)) = 1$

$((Y:=Y-1, X:=2*X), \sigma_{\substack{X\mapsto 2\\ Y\mapsto 2}}) \Downarrow \sigma_{X\mapsto 4, Y\mapsto 1}$

$(n, \sigma_{\substack{X\mapsto 2\\ Y\mapsto 2}}) \Downarrow \sigma_{X\mapsto 4, Y\mapsto 1}$

$e_\sigma(\neg(Y=0)) = 1$

$((X:=2*X, Y:=Y-1), \sigma_{\substack{X\mapsto 4\\ Y\mapsto 1}}) \Downarrow \sigma_{X\mapsto 8, Y\mapsto 0}$

$(n, \sigma_{\substack{X\mapsto 4\\ Y\mapsto 1}}) \Downarrow \sigma_{X\mapsto 8, Y\mapsto 0}$ (regula 2 - while)

S.S.

$(n, \sigma_{X\mapsto 1, Y\mapsto 3}) \longrightarrow$ (if $\neg(Y=0)$ then $Y:=Y-1, X:=2*X$; n else skip, $\sigma_{X\mapsto 1, Y\mapsto 3}) \longrightarrow ((Y:=Y-1, X:=2*X), n, \sigma_{X\mapsto 1, Y\mapsto 3}) \rightarrow$

$\rightarrow$ (skip, $X:=2*X, \sigma_{X\mapsto 1, Y\mapsto 2}) \rightarrow (X:=2*X, \sigma_{X\mapsto 1, Y\mapsto 2}) \rightarrow$

$\rightarrow$ (skip, $\sigma_{X\mapsto 2, Y\mapsto 2}) \longrightarrow (n, \sigma_{X\mapsto 2, Y\mapsto 2}) \longrightarrow$ (if $\neg(Y=0)$ then $Y:Y-1, X:=2*X$; n else skip, $\sigma_{X\mapsto 2, Y\mapsto 2}) \rightarrow ((Y:=Y-1, X:2*X), n, \sigma_{X\mapsto 2, Y\mapsto 2}) \rightarrow$ (skip, n, $X=2*X, \sigma_{X\mapsto 2, Y\mapsto 1}) \longrightarrow (n, X:=2*X, \sigma_{X\mapsto 2, Y\mapsto 1}) \rightarrow$ (skip, $n, \sigma_{X\mapsto 4, Y\mapsto 1}) \longrightarrow (n, \sigma_{X\mapsto 4, Y\mapsto 1}) \rightarrow$ -----

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2) Programare logică:

- x, y, z, u, v, w variabile
- a, b, c constante
- h, g fct. de aritate 1
- f fct. de aritate 2
- p fct. de aritate 3

2.1) UNIFICARE:

1) $f(x, y) = f(h(x), x)$; $f(h(x), x) = f(x, b)$

Sol.: $f(x, y) = f(h(x), x)$; $f(h(x), x) = f(x, b)$ | $\emptyset$

$x = h(x)$; $y = x$; $h(x) = x$; $x = b$

EȘEC ($x \in Var(h(x))$

2) $f(x, f(x, g(y))) = f(u, z)$; $f(u, z) = f(g(y), y)$ | $\emptyset$

$x = u$; $f(x, g(y)) = z$; $u = g(y)$; $z = y$ | $z = y$

$x = u$; $f(x, g(y)) = z$; $u = g(y)$ | $x = u$

$f(u, g(y)) = z$; $u = g(y)$ | $u = g(y)$

$f(g(y), g(y)) = y$ | $\emptyset$

$y = f(g(y), g(y))$

EȘEC ($y \in Var(f(g(y), g(y))$)

3) Sol.: $f(f(x, y), x) = f(g(y), z)$; $f(g(y), z) = f(u, h(z))$ | $\emptyset$

$f(x, y) = g(y)$; $x = z$; $g(y) = u$; $z = h(z)$ | $x = z$

$f(z, y) = g(y)$; $g(y) = u$; $z = h(z)$

EȘEC ($f \neq g$, adică fct. diferite)

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4) $f(f(x,y),x) = f(v,u)$; $f(v,u) = f(u,h(z))$

$f(x,y) = v$; $\underline{x = u}$; $v = u$; $u = h(z)$

$f(u,y) = v$; $\underline{v = u}$; $u = h(z)$

$f(u,y) = u$; $\underline{u = h(z)}$;

$f(h(z),y) = h(z)$

$\emptyset$

$x = u$

$v = u$

$u = h(z)$

EȘEC ($f \neq h$ / SAU, dacă nu mai facem ultimul pas, $u \in Val_2 f(u,y)$)

5) $f(f(x,y),x) = f(v,u)$; $f(v,u) = f(u,z)$

$f(x,y) = v$; $x = u$; $v = u$; $\underline{u = z}$

$f(x,y) = v$; $\underline{x = z}$; $v = z$

$f(z,y) = v$; $\underline{v = z}$

$f(z,y) = z$;

$z = f(z,y)$

EȘEC ($z \in Val_2(f(z,y))$)

$\emptyset$

$u = z$

$x = z$

$v = z$

$\emptyset$

6) $f(f(g(x),h(y)),h(z)) = f(f(u,h(h(x))),h(y))$; $f(f(u,h(h(x))),h(y)) = f(v,w)$

$f(g(x),h(y)) = f(u,h(h(x)))$; $h(z) = h(y)$; $f(u,h(h(x))) = v$; $h(y) = w$

$g(x) = u$; $h(y) = h(h(x))$; $z = y$; $f(u,h(h(x))) = v$; $h(y) = w$

$g(x) = u$; $y = h(x)$; $\underline{z = y}$; $f(u,h(h(x))) = v$; $h(y) = w$

$\emptyset$

$\emptyset$

$\emptyset$

$z = y$

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$g(x)=u$; $y=h(x)$; $f(u,h(h(x)))=v$; $h(y)=w$

$g(x)=u$; $f(u,h(h(x)))=v$; $h(h(x))=w$

$u=g(x)$; $v=f(u,h(h(x)))$; $w=h(h(x))$

$u=g(x)$; $v=f(u,h(h(x)))$

$v=f(g(x),h(h(x)))$

$y=h(x)$

$\emptyset$

$w=h(h(x))$

$u=g(x)$

$v=f(g(x),h(h(x)))$

SOLUȚIA: $z=h(x)$; $y=h(x)$; $w=h(h(x))$;

$u=g(x)$; $v=f(g(x)\,h(h(x)))$

7) $p(x,x,z)=p(f(a,a),y,y)$; $p(f(a,a),y,y)=p(f(x,a),b,x)$

$x=f(a,a)$; $x=y$; $z=y$; $f(a,a)=f(x,a)$; $y=b$; $y=z$

$x=f(a,a)$; $x=y$; $z=y$; $a=x$; $a=a$; $y=b$

$x=f(a,a)$; $x=b$; $z=b$; $a=x$

$x=f(a,a)$; $x=b$; $z=b$; $x=a$

$a=f(a,a)$; $a=b$; $z=b$

$\emptyset$

$\emptyset$

$y=b$

$\emptyset$

$x=a$

EȘEC ($a\in Var(f(a,a))$)

8) $p(x,x,z)=p(f(a,a),y,y)$; $p(f(a,a),y,y)=p(x,b,z)$

$x=f(a,a)$; $x=y$; $z=y$; $f(a,a)=x$; $y=b$; $y=z$

$x=f(a,a)$; $x=b$; $z=b$; $f(a,a)=x$; $b=z$

$x=f(a,a)$; $x=b$

$b=f(a,a)$

EȘEC ($b\neq f\Rightarrow$ { b e fct. de aritate 0 (constantă); f e fct. de aritate 2)

$\emptyset$

$y=b$

$z=b$

$x=b$

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9) $h(x,x,z) = h(f(a,a),y,y)$; $h(f(a,a),y,y) = h(x,f(a,a),z)$ | $\emptyset$

$x = f(a,a)$; $x = y$; $z = y$; $f(a,a) = x$; $\underline{y = f(a,a)}$; $y = z$ | $y = f(a,a)$

$x = f(a,a)$; $x = f(a,a)$; $z = f(a,a)$; $f(a,a) = z$

SOLUȚIE: $x = f(a,a)$; $y = f(a,a)$; $z = f(a,a)$

10) $h(f(x,a),g(y),z) = h(f(a,a),z,u)$; $h(f(a,a),z,u) = h(v,u,z)$ | $\emptyset$

$f(x,a) = f(a,a)$; $g(y) = z$; $z = u$; $f(a,a) = v$; $z = u$; $u = z$ | $\emptyset$

$x = a$; $a = a$; $z = g(y)$; $z = u$; $\underline{v = f(a,a)}$ | $v = f(a,a)$

$\underline{x = a}$; $z = g(y)$; $z = u$ | $x = a$

$\underline{z = g(y)}$; $z = u$ | $z = g(y)$

$g(y) = u$ | $\emptyset$

$u = g(y)$

SOLUȚIE: $x = a$; $v = f(a,a)$; $z = g(y)$; $u = g(y)$

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2.2) REZOLUȚIE:

1) $x: -R, Q.$ $t.$ $?-w.$

$S: -R, Q.$ $Q.$

$v: -t, u.$ $u.$

$w: -v, S.$ $R.$

Soluție: $G_0 = \neg w$

$C_0 = w \vee \neg v \vee \neg S$

$G_1 = \neg v \vee \neg S$

$C_1 = v \vee \neg t \vee \neg u$

$G_2 = \neg S \vee \neg t \vee \neg u$

$C_2 = t$

$G_3 = \neg S \vee \neg u$

$C_3 = u$

$G_4 = \neg S$

$C_4 = S \vee \neg R \vee \neg Q$

$G_5 = \neg R \vee \neg Q$

$C_5 = R$

$G_6 = \neg Q \longrightarrow C_6 = Q \longrightarrow \boxed{G_7 = \bot}$

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2) $g(x,y) :- g(y,x), g(y, f(f(y)))$.

$g(a, f(f(x)))$.

$? - g(f(z), a)$.

$C_0 = \neg g(f(z), a)$

$C_0 = g(f(z), a) \vee \neg g(a, f(z)) \vee \neg g(a, f(f(a)))$ | $\begin{cases} y = a \\ x = f(z) = f(f(a)) \end{cases}$

$C_1 = \neg g(a, f(z)) \vee \neg g(a, f(f(a)))$ | $\begin{cases} z = f(x') = f(a) \end{cases}$

$C_1 = g(a, f(f(x')))$ |

$C_2 = \neg g(a, f(f(a)))$ | $\begin{cases} x' = a \end{cases}$

$C_2 = g(a, f(f(x')))$ |

$C_3 = \perp$ |

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3) $p(x) :- q(x, f(y)), r(a).$

$p(x) :- r(x).$

$q(x,y) :- p(y).$

$r(x) :- q(x,y).$

$r(f(b)).$

$? - p(x), q(y,z).$

$C_0 = \neg p(x) \vee \neg q(y,z)$ |

$C_0 = p(x') \vee \neg r(x')$ | $\{x = x'$

$C_1 = \neg q(y,z) \vee \neg r(x')$ |

$C_1 = q(y',z') \vee \neg p(z')$ | $\begin{cases} y = y' \\ z = z' \end{cases}$

$C_2 = \neg r(x') \vee \neg p(z')$ |

$C_2 = r(f(b))$ | $\{x' = f(b)$

$C_3 = \neg p(z')$ |

$C_3 = p(w) \vee \neg r(w)$ | $\{z' = w$

$C_4 = \neg r(w)$ |

$C_4 = r(f(b))$ | $\{w = f(b)$

$C_5 = \perp$ |

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3) Programare funcțională:

3.1) LAMBDA-CALCUL FĂRĂ TIPURI:

1) $((\lambda z.z)(\lambda q.(qq)))(\lambda s.(sa)) \longrightarrow_\beta$

$(\lambda q.(qq))(\lambda s.(sa)) \longrightarrow_\beta$

$(\lambda s.(sa))(\lambda s.(sa)) \longrightarrow_\beta$

$(\lambda s.(sa))\ a \longrightarrow_\beta$

$a\ a$

2) $((\lambda z.z)(\lambda z.(zz))(\lambda z.(zq)) \longrightarrow_\beta$

$(\lambda z.(zz))(\lambda z.(zq)) \longrightarrow_\beta$

$(\lambda z.(zq))(\lambda z.(zq)) \longrightarrow_\beta$

$(\lambda z.(zq))\ q \longrightarrow_\beta$

$q\ q$

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3) $((\lambda s.\lambda z.(s\,z\,z))(\lambda a.a))\,b \longrightarrow_\beta$

$(\lambda z.(\lambda a.a)\,z\,z)\,b \longrightarrow_\beta$

$((\lambda a.a)(b))b \longrightarrow_\beta$

$b\ b$

4) $(((\lambda s.\lambda z.(s\,z\,z))(\lambda z.z))\,z \longrightarrow_\beta$

$(\lambda z.(\lambda z'.z')\,z\,z)\,z'' \longrightarrow_\beta$

$((\lambda z'.z')\,z'')\,z'' \longrightarrow_\beta$

$z''\ z''$

5) $((\lambda s.(s\,s))(\lambda z.z))(\lambda z.z) \longrightarrow_\beta$

$((\lambda z.z)(\lambda z.z))(\lambda z.z) \longrightarrow_\beta$

$(\lambda z.z)(\lambda z.z) \longrightarrow_\beta$

$\lambda z.z$

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3.2) LAMBDA-CALCUL CU TIPURI:

1) $\lambda xyz.(x(yz)) \Rightarrow \lambda x.\lambda y.\lambda z.(x(yz))$

$x : X$; $y : Y$; $z : Z$

$Y = Z \to Y'$

$X = Y' \to X'$

$t \overset{not.}{=} \lambda xyz.(x(yz)) = (Y' \to X') \to (Z' \to Y') \to Z \to X'$

$$\frac{}{\Gamma \vdash x : Y' \to X'} \qquad \frac{\Gamma \vdash y : Z \to Y' \quad \Gamma \vdash z : Z}{\Gamma \vdash yz : Y'}$$

Γ // not.

$\{x : Y' \to X', y : Z' \to Y', z : Z\} \vdash (x(yz)) : X'$

$\{x : Y' \to X', y : Z' \to Y'\} \vdash \lambda z.(x(yz)) : Z \to X'$

$\{x : Y' \to X'\} \vdash \lambda yz.(x(yz)) : (Z' \to Y') \to Z \to X'$

$\emptyset \vdash \lambda xyz.(x(yz)) : (Y' \to X') \to (Z' \to Y') \to Z \to X'$

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2) $\lambda xy.(xy(\lambda z.y)) \iff \lambda x.\lambda y.(xy(\lambda z.y))$

$x : X \;;\; y : Y \;;\; z : Z$

$\lambda xy.\underbrace{xy}_{Y}\underbrace{(\lambda z.y)}_{Z \to Y} \Rightarrow X : Y \to (Z \to Y) \to W$

$$\frac{\Gamma \vdash x : Y \to (Z \to Y) \to W \quad \Gamma \vdash y : Y}{\Gamma \vdash xy : (Z \to Y) \to W} \qquad \frac{\Gamma \cup \{z : Z\} \vdash y : Y}{\Gamma \vdash \lambda z.y : Z \to Y}$$

$\overset{not.}{=\!=} \Gamma$

$\{x : Y \to (Z \to Y) \to W, y : Y\} \vdash (xy)(\lambda z.y) : W$

$\{x : Y \to (Z \to Y) \to W\} \vdash \lambda y.xy(\lambda z.y) : Y \to W$

$\emptyset \vdash t : (Y \to (Z \to Y) \to W) \to Y \to W$

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3) $(\underbrace{\lambda xyz.zxy}_{x\to y\to z\to w})(\overbrace{\lambda xyz.y}^{x'\to y'\to z'\to y''})(\underbrace{\lambda xy.y}_{x''\to y''\to y''})$

$z: x \to y \to w$

$z: (x' \to y' \to z' \to y') \to (x'' \to y'' \to y'') \to w$

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1.2) SEMANTICĂ AXIOMATICĂ:

1) $\{X=n \wedge Y=m\}(x:=x+y; y:=x-y); x:=x-y\{x=m \wedge y=n\}$

$\{X=n \wedge Y=m\}\ x:=x+y\{x=n+m \wedge y=m\}$

$\{x=n+m \wedge Y=m\}\ y:=x-y\{x=n+m \wedge y=n\}$

$\{x=n \wedge Y=m\}\ x:=x+y;\ y:=x-y\{x=n+m \wedge y=n\}$

$\{x=n+m \wedge y=n\}\ x:=x-y\{x=m \wedge y=n\}$

2) $\{1 \leq N\}(P:=0; C:=1);$ while $C \leq N$ do $(P:=P+M; C:=C+1)\{P=M*N\}$

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